Data Operations and Merging Tables

DATA SCIENCE BOOTCAMP

NASHVILLE SOFTWARE SCHOOL

Objectives

1. Learn about the basic kinds of operations that can be performed on tabular data.
2. Learn the different types of merges.

Data Operations

- Inspection
  - shape
  - head/tail
  - info
  - sort_values

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  - sort_values
- Modification
  - renaming columns
  - dropping or adding columns
  - dropping or filling NaNs
  - dropping duplicates

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  - shape
  - head/tail
  - info
  - sort_values
- Modification
  - renaming columns
  - dropping or adding columns
  - dropping or filling NaNs
  - dropping duplicates
- Selection
  - Slicing: selecting columns
  - Filtering: selecting rows
  - loc/iloc/[ ]

Data Operations

- Summarization/Aggregation
  - value_counts
  - sum/mean/count/median/min/max
  - describe
  - groupby + aggregation

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  - melt (wide to long)

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  - value_counts
  - sum/mean/count/median/min/max
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- Reshaping
  - pivot/pivot_table (long to wide)
  - melt (wide to long)
- Combining Tables
  - Merging
  - Concatenating

Merging

A merge or join is used to combine data from two tables.

mlb

City	MLB_Team
Atlanta	Braves
Detroit	Tigers
Houston	Astros
Milwaukee	Brewers
Toronto	Blue Jays

nfl

City	NFL_Team
Atlanta	Falcons
Charlotte	Panthers
Detroit	Lions
Houston	Texans
Nashville	Titans

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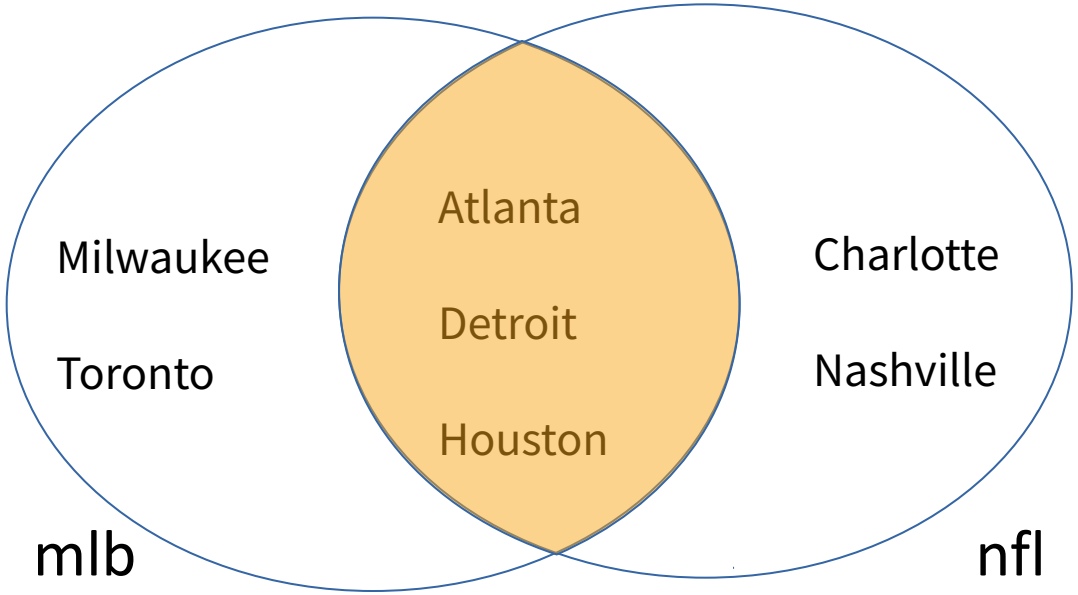
Inner Joins

Selects all indices appearing in table 1 AND table 2.



Inner Joins

City	MLB_Team
Atlanta	Braves
Detroit	Tigers
Houston	Astros
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City	NFL_Team
Atlanta	Falcons
Charlotte	Panthers
Detroit	Lions
Houston	Texans
Nashville	Titans

pd.merge(mlb, nfl) OR pd.merge(mlb, nfl, how = ‘inner’)
OR pd.merge(mlb, nfl, on = ‘City’)

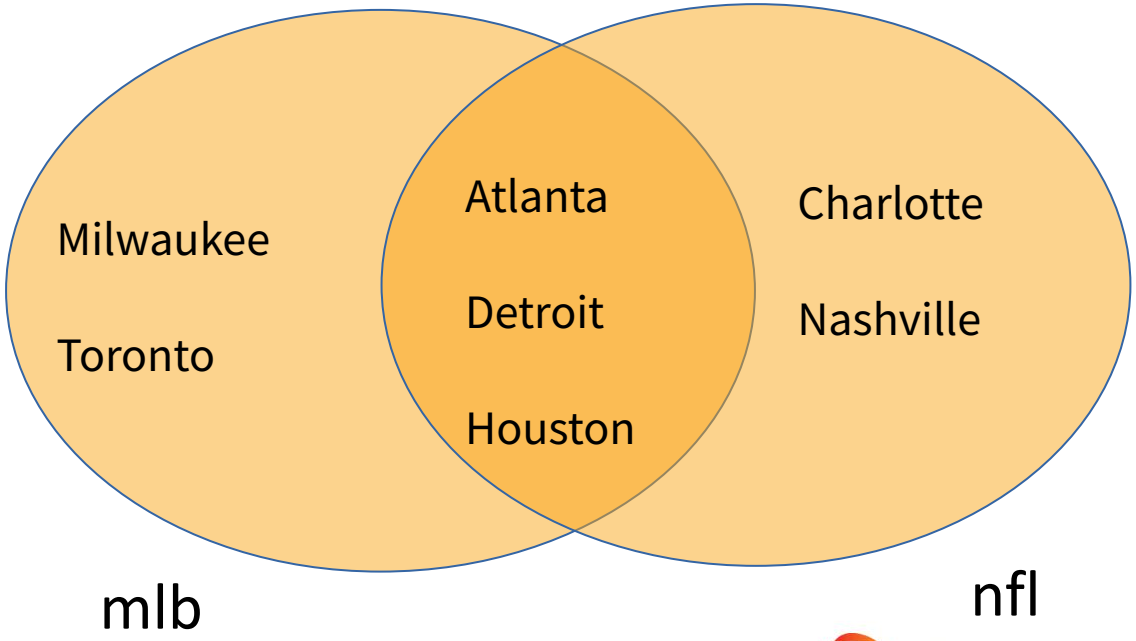
City	MLB_Team	NFL_Team
Atlanta	Braves	Falcons
Detroit	Tigers	Lions
Houston	Astros	Texans

Joins in Pandas

- By default, pd.merge() will join on all columns that appear in both tables (with the exact same name)
- The exact column(s) to join on can be specified by passing a list to the ‘on =’ argument.
- If the column to join on has a different name in the different tables, it can be specified with a ‘left_on =’ and a ‘right_on =’ argument.

Outer Join

Selects all indices appearing in table 1 OR table 2.
Rows without a match are filled with NaN.



Outer Join

City	MLB_Team
Atlanta	Braves
Detroit	Tigers
Houston	Astros
Milwaukee	Brewers
Toronto	Blue Jays

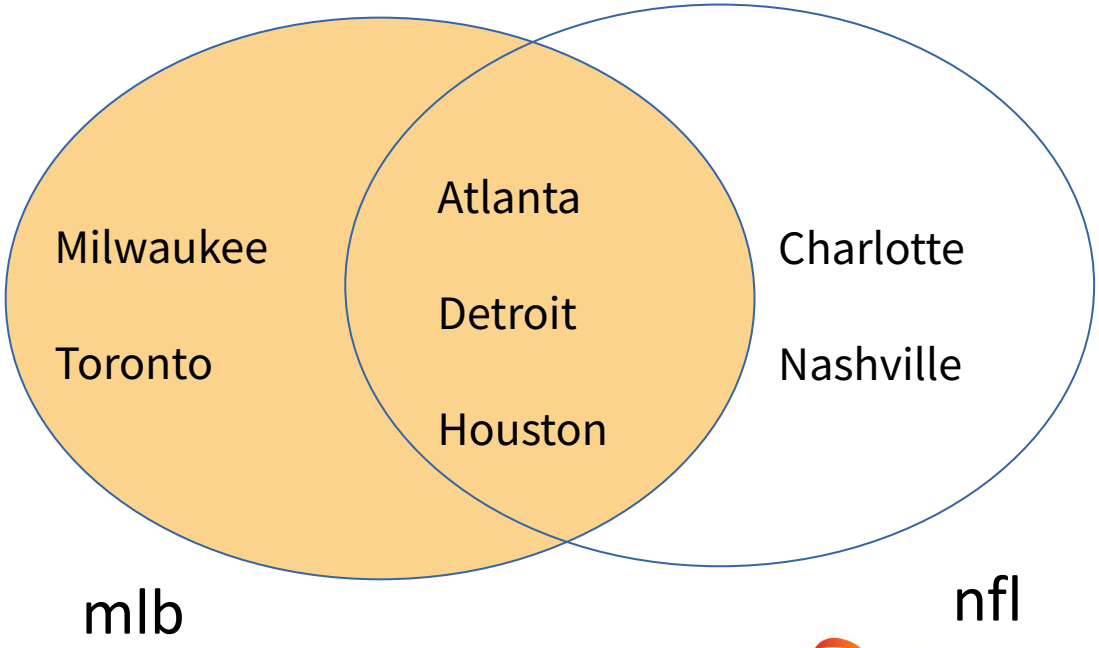
City	NFL_Team
Atlanta	Falcons
Charlotte	Panthers
Detroit	Lions
Houston	Texans
Nashville	Titans

pd.merge(mlb, nfl, how = ‘outer’)

City	MLB_Team	NFL_Team
Atlanta	Braves	Falcons
Detroit	Tigers	Lions
Houston	Astros	Texans
Milwaukee	Brewers	**NaN**
Toronto	Blue Jays	**NaN**
Charlotte	**NaN**	Panthers
Nashville	**NaN**	Titans

Left (Outer) Join

Selects all indices in the left table and any matching indices from the right table.
Rows from the left table without a match are filled with NaN.



Left (Outer) Join

City	MLB_Team
Atlanta	Braves
Detroit	Tigers
Houston	Astros
Milwaukee	Brewers
Toronto	Blue Jays

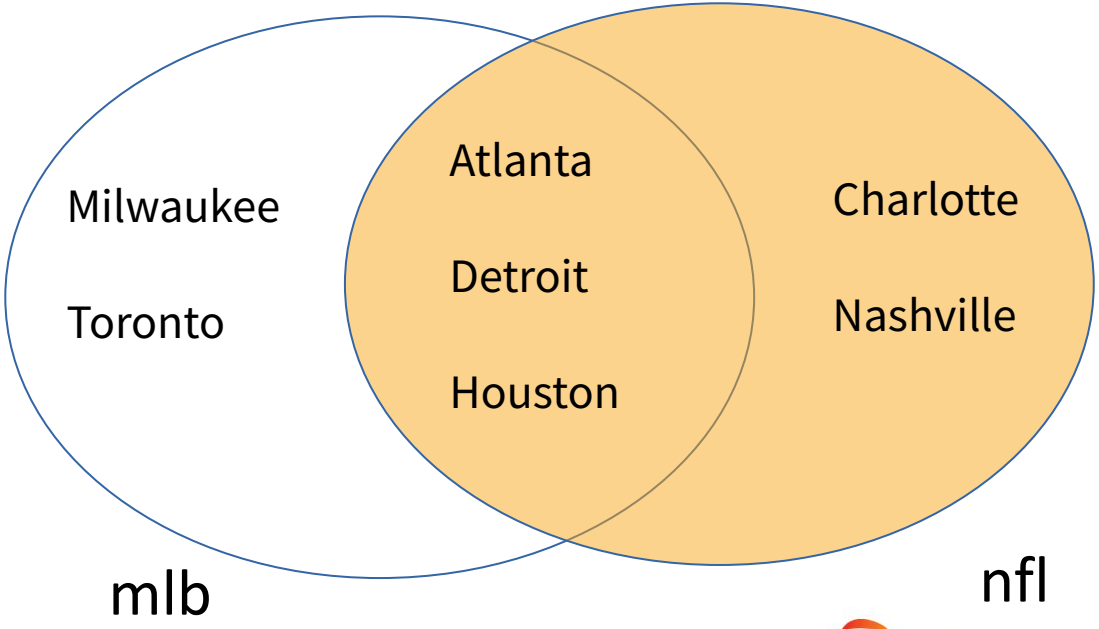
City	NFL_Team
Atlanta	Falcons
Charlotte	Panthers
Detroit	Lions
Houston	Texans
Nashville	Titans

```
pd.merge(mlb, nfl, how = 'left')
```

City	MLB_Team	NFL_Team
Atlanta	Braves	Falcons
Detroit	Tigers	Lions
Houston	Astros	Texans
Milwaukee	Brewers	**NaN**
Toronto	Blue Jays	**NaN**

Right (Outer) Join

Selects all indices in the right table and any matching indices from the left table.
Rows from the right table without a match are filled with NaN.



Right (Outer) Join

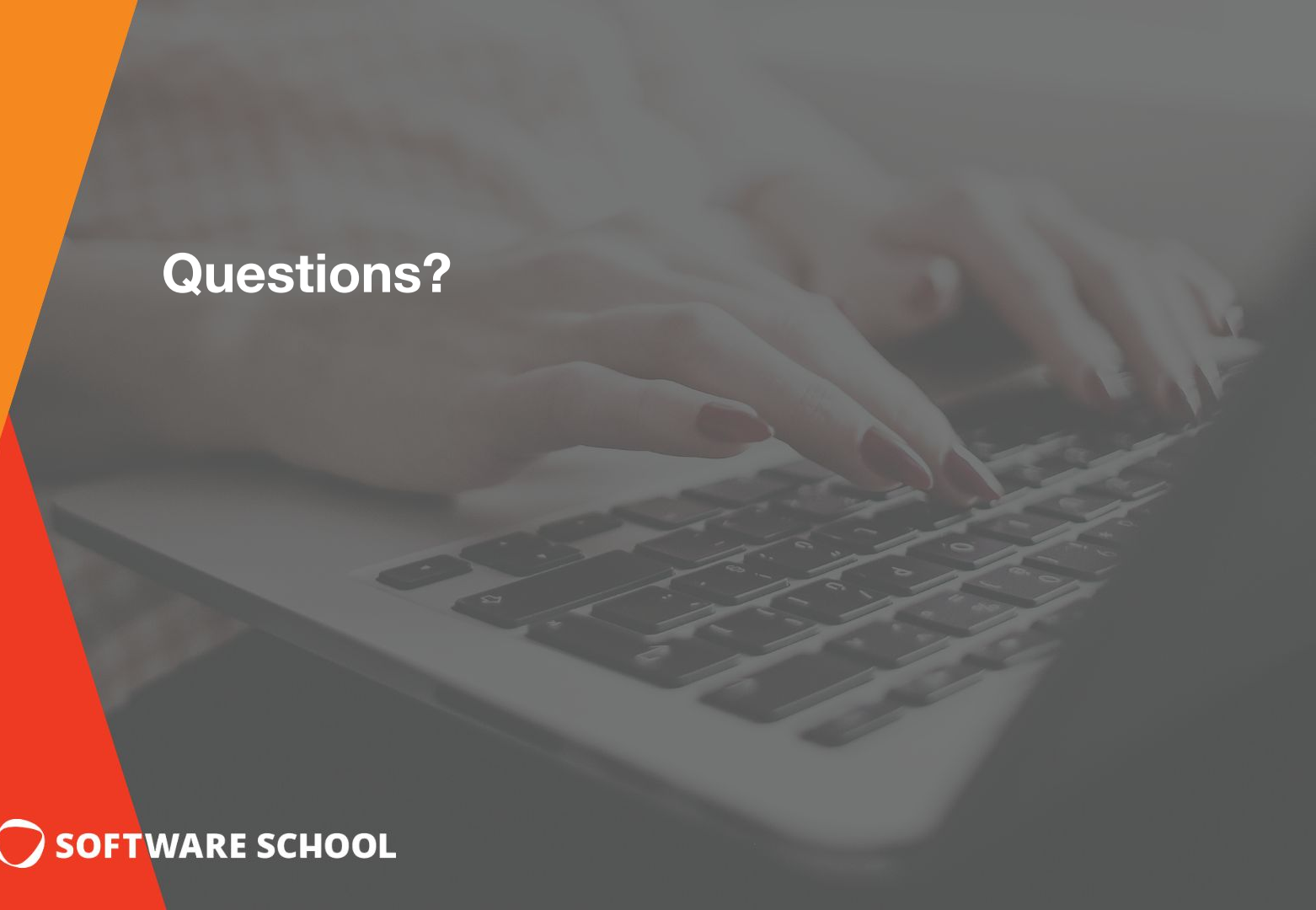
City	MLB_Team
Atlanta	Braves
Detroit	Tigers
Houston	Astros
Milwaukee	Brewers
Toronto	Blue Jays

City	NFL_Team
Atlanta	Falcons
Charlotte	Panthers
Detroit	Lions
Houston	Texans
Nashville	Titans

```
pd.merge(mlb, nfl, how = 'right')
```

City	MLB_Team	NFL_Team
Atlanta	Braves	Falcons
Detroit	Tigers	Lions
Houston	Astros	Texans
Charlotte	**NaN**	Panthers
Nashville	**NaN**	Titans

Questions?



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